

Microprocessor Simulators

Simulation software apply mathematical models to replicate the behaviour of an actual electronic hardware. This month's EFY-CD includes simulators to test codes developed for microprocessors

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Microprocessors are semiconductor devices used for general-purpose computing. These include an arithmetic logic unit, register arrays and control circuits on a single chip. A microprocessor is used along with various devices like memory and input and output devices to perform a particular task or function. Microprocessor simulators are software that help developers to test their code development for microprocessors without using the actual hardware.

In this month's EFY-CD, we have included some software for microprocessor simulation. These simulation tools apply mathematical models to replicate the behaviour of an actual electronic hardware. Using them you can make your computer a complete lab for testing codes developed for microprocessors. Try your hands on these tools to explore more.



An open source software development tool for the Atmel AVR series of RISC microprocessors.

What's Special?

This is basically a collection of tools that comprises a compiler, a programmer and a debugger and works on Windows platform. The setup includes the GNU GCC compiler for C and C++. This easy-to-use application is useful for everyone, starting from hobbyists and school students to commercial project developers. Further, it has AVaRIC, which connects the JTAG ICE to the AVR-GDB debugging software, a makefile editor called 'MFile' and

SRecord tool for editing EPROM load files.

Distributor: Being free and open source, this software can be downloaded freely from the website.

*Home Page: <http://winavr.sourceforge.net/>
Download URL: <http://winavr.sourceforge.net/download.html>*

Platform Supported: Windows



CEDAR LS is an interactive digital logic simulator.

What's Special?

This feature-rich application comes with a user interface that is built with wxWidgets and is used for teaching the logic design or testing simple digital designs. It features both low-level logic gates as well as high-level components, including registers and a Z80 microprocessor emulator. This application comes with a user-friendly circuit navigation that supports drag-and-drop and interactive wiring system.

Distributor: Being free and open source, this software can be downloaded freely from the website.

Home Page: <http://cedarlogic.scienceontheweb.net>

Download URL: <http://sourceforge.net/projects/cedarlogic/files/>

Platform Supported: Windows

Microprocessor Simulator



This simulator is for low-level programming and microcomputer architecture. It can be used by any student studying low-level programming, control or machine architecture for the first time.

What's Special?

The simulator emulates an eight-bit CPU that is similar to the low eight bits of the 80x86 family of chips. It includes a small sub-set of the full instruction set normally found with this style of processor. It also includes advanced instructions such as CALL, RET, INT and IRET. There is a hardware timer interrupt simulation too.

Distributor: Being free and open source, this software can be downloaded freely from the website.

Home Page: www.softwareforeducation.com

Download URL: www.softwareforeducation.com/sms32v50/

Platform Supported: Windows



PTLsim is a cycle-accurate x86 microprocessor simulator and virtual machine for the x86 and x86-64 instruction sets.

What's Special?

PTLsim is very different from most cycle-accurate simulators. Because it runs directly on the platform that it is simulating, it is able to switch in and out of full out-of-order simulation mode and native x86 or x86-64 mode at any time completely transparent to the running user code. It supports the full x86-64 instruction set of the Pentium 4+, Athlon 64 and similar machines with all extensions.

Distributor: Being free and open source, this software can be downloaded freely from the website.

Home Page: www.ptlsim.org/

Download URL: www.ptlsim.org/download.php

Platform Supported: Linux

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